

First Course Handout for PHY597/599/697/699

Academic Year: 2024–2025, 2nd Semester

- 1) **Objectives and course content:** The main objective of the course is to let the students do a well-defined new research problem. It is also important that the student should be able to communicate her/his work in the form of a short talk in front of the course-committee. Additionally, the students are expected to learn from their respective project-supervisors how to write their scientific works in the form of a respectable scientific journal paper.
- 2) **Evaluation Components & Policies: Exams, Quizzes, Assignments, Attendance, Participation etc.:**
 - A) Strictly by **14 January 2025** the students should submit a short write-up (maximum one page) mentioning the tentative title, the supervisor(s) and the tentative plan of the project. Those who will be continuing the project done in the preceding semester **must** add a paragraph detailing what has already been done in the project. In case the write-up is not received, then it will be assumed that corresponding student intends to drop the course and action will be initiated to deregister the student from the course.
 - B) A mid-semester report detailing the progress and the difficulties (if any) and directions for their resolution have to be submitted on **01 March 2025**. There is no fixed format but a word count of about 500 words is preferred. Additionally, either a short presentation in front of entire committee or an in person short discussion with one of the committee members will be scheduled tentatively on the day.
 - C) An end-semester report should be submitted in the form of a complete paper on **30 April 2025**. Maximum length: 6 pages (including figures). Anything written beyond the sixth page will **not** be evaluated. The format must strictly be that of **Physical Review A/B/C/D/E**.
 - D) Hardcopies of all the reports/write-ups mentioned above should be dropped into **Sagar Chakraborty's mailbox** in front of the physics office and **also a soft-copy** (that need not bear supervisor's signature) should be emailed. Any hardcopy of the reports without the corresponding supervisor's signature will **not** be considered. In case of multiple supervisors, it **must** be made sure that at least one supervisor is from the Department of Physics, IIT Kanpur: Only that supervisor will be formally responsible for the supervisors' component of the marks to be given to her/his project students.
 - E) An end semester presentation (ppt or equivalent) in offline mode on **07 May 2025** is mandatory. The slides of the presentations should be nicely and

professionally prepared. Duration of each talk will be of **12+3 minutes** (it includes the time required for questions and answers).

F) Marks Distribution:

Supervisor: 50 Marks

- (i) Regularity and interactions (till mid-sem): 10 marks
- (ii) Workwise progress (till mid-sem): 10 marks
- (iii) Regularity and interactions (mid-sem to end-sem): 10 marks
- (iv) Workwise progress (mid-sem to end-sem): 10 marks
- (v) Scientific writing skill (project reports): 10 marks

Committee: 50 Marks

- (i) Project reports: 20 marks
- (ii) Seminar: 30 marks (Slide-organization and adherence to time-limit: 10, Overall clarity of talk and communication skills: 10, Questions & answers: 10)

4) **Course Policies: Attendance, Honesty Practices, Withdrawal**

- A) It is the responsibility of the students (and not the supervisors or the instructor-in-charge) to seek and maintain regular discussion sessions with their supervisors.
- B) The submission of the end-semester report and deliverance of the end-semester presentation are mandatory. Missing any one of these will straightaway fetch the student an **F grade**.
- C) Any kind of plagiarism or any other form of unfair means at any stage of the project work will be very strictly dealt with.
- D) Missing any of the deadlines will strictly mean **zero marks** in the corresponding component of evaluation.
- F) If required, the students should not hesitate at all to contact the instructor-in-charge, Sagar Chakraborty (mobile: 8601953677; email: sagarc@iitk.ac.in), at any point of time.